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A hybrid XGBoost–SVM ensemble framework for robust cyber-attack detection in the internet of medical things (IoMT)

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Today, the rise of the Internet of Medical Things (IoMT) has evolved into a highly valued global market worth billions of dollars. However, this growth has also created many opportunities for massive and advanced attack scenarios due to the vast number of devices and their interconnected communication networks. Based on recent reports, it is observed that during the Covid-19 pandemic, the necessity of the IoMT ecosystem has increased significantly. On the other hand, attackers and intruders aim to impair data integrity and patient safety with the prevalence of sophisticated cyber attacks including Man in the Middle (MITM) attacks like spoofing and data injection. In this research work, WUSTL-EHMS-2020 dataset is utilized to demonstrate a robust IoMT cyberattack detection method based on machine learning and the efficiency of the proposed model is validated by employing TON-IoT and CICIDS 2017 datasets. We offer an ensemble approach that employs Extreme Gradient Boosting (XGBoost) and Support Vector Machine (SVM) classifiers to address the exclusive challenges in IoMT security. By utilizing the complementing advantages of SVM's decision boundary precision and XGBoost's gradient-based optimization, our model outperforms baseline techniques with a superior detection accuracy of 98.04% with WUSTL-EHMS-2020 dataset. In addition, the practicality of the proposed model is examined by considering peculiar features of IoMT like resource restrictions, medical device communication diversity, and healthcare data privacy by comparing with IoMT and IoT datasets by exposing the patterns of cyberattack in dynamic IoMT environment. Hence, this research will be considered as the pioneer for developing reliable IoMT security solutions by adapting trustworthy and scalability.

Keywords IoMT, Machine learning, XGBoost, SVM, Soft voting, Cyber attacks

The worldwide impact after COVID-19 pandemic period is devastating and incomparable which reformed the economy capacity of IoMT (Internet of Medical Things) to 41.17 billion USD in 2020. In 2028, it will grasp approximately 187.60 billion USD with Compound annual growth rate (CAGR) of 29.5% on the basis of recent forecast analysis. It demonstrates the significant contribution of IoMT in enhancing the market size of intelligent health systems which is implemented in several extents like Mobile Health, IoT-based healthcare systems and AI-powered Healthcare for monitoring patients remotely and effective medicine management.

Though IoMT is derived from IoT (Internet of Things), it built with the important components like medical services, healthcare devices and software applications. IoMT real time data is distributed using network communication technologies such as ZigBee, Wi-Fi, Near Field Communication (NFC), Bluetooth and Local Area Network (LAN). Message Queue Telemetry Transport (MQTT) protocol is generally used by IoMT. IoMT network and its key components are illustrated in Fig. 1.

The subsequent facts define the necessity of IoMT. After the evolution of IoMT, remote patient monitoring emerged technically due to the vast increment of long-lasting diseases and elder persons. United Nations

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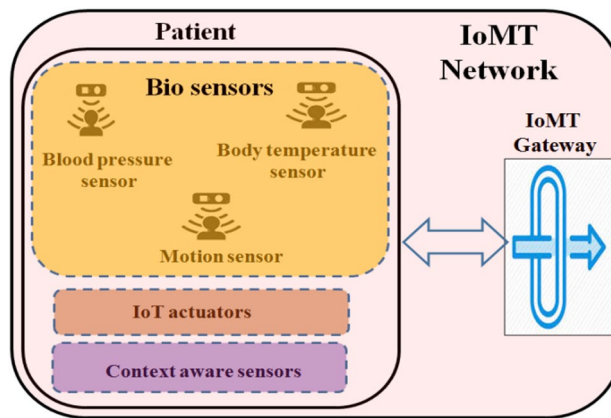


Fig. 1. IoMT Network.

estimates that the count of aged people will reach to 2.1 billion globally in 2050. Health issues are resolved by integrating IoMT devices with mobile applications which reduced the lack in connecting patients with physicians by ensuring secure health data sharing. In the period of worldwide Covid-19 pandemic, it helps patients in managing needless hospital visits.

According to Goldman Sachs reports, IoMT played key role in reducing the cost of health care business and annual healthcare expenditures in USA by 300 billion USD. Consecutive pandemic periods of Covid 19 enflamed healthcare awareness of each human and increased the need of E-Health systems and devices.

As IoMT is associated with enormous amount of healthcare gadgets as well as sensitive information related to patients, hospitals and doctors¹. Such critical confidential data is stored in data centres and shared to users through gateways. Further, IoMT extended the utilization of IoT appliances in various fields of medical industry by incorporating medical sensors, gadgets, automated robots and UAVs (Unmanned Aerial Vehicles). Medical sensors and actuators are used as wearable gadgets while considering BANs (Body Area Networks). These smart technologies improved health monitoring by effectively observing physical conditions of patients. Other side, medical robots are utilised for minor surgical operations.

However, main issue of various IoMT devices is insecurity and they can be infected by the vulnerable activities of attackers. Security breaches in the IoMT ecosystem will cause danger to the lives of patients. Attackers are targeting confidential data of the system by breaking data servers. Massive attack vectors can be exploited by attackers in smart medical appliances like smart pacemakers too which is highly dangerous. It is mandatory to care about the security exploitations and vulnerabilities and protect the IoMT healthcare ecosystem in secured manner for ensuring sound healthcare environment with zero risks².

Based on IBM survey it is observed that huge losses are acquired for healthcare industries because of data breaches³. According to SonicWall cyber threat report, ransomware attack occurrence count of healthcare industry in 2020 is hiked to 123% during Covid-19 pandemic period. Help Net Security reported that attackers broke health system of Singapore for obtaining individual information of 1.5 million persons including the prime minister of the country.

Various reasons of multiple IoMT related cyber attacks are discovered and detailed⁴. Security and privacy issues of IoMT ecosystem are identified such as Compatibility and complexity risks, application risks and CIA (confidentiality, integrity, and availability) impair. An insecure IoMT ecosystem can't hold patients record securely. Attackers eagerly focused on healthcare data by implementing MIM attacks, data theft, probe attacks, decrypted network communication and DoS/DDoS attacks. These security assaults should be avoided to protect medical devices and system.

Malicious attackers targeted healthcare industry continuously with the implementation of security exploitation and massive attacks. Attackers weaken the integrity of IoMT ecosystem and target patient's confidential information by applying DDoS attack to damage the system. These kinds of context imply that it is highly essential to ensure the security of IoMT ecosystem by applying advanced technologies and combating against intruders/attackers.

Research works are carried in order to ensure security of smart medical appliances and wearable BAN devices by applying cryptographic methods such as encryption, authentication and trust related systems. Further physical layer security is also considered to improve security in IoMT devices⁵. This conventional information technology has many challenges to apply and it appropriate security assurance is difficult. Therefore in addition physical layer security risk factors of communication media should be addressed too.

The key contribution of this paper is summarized as follows:

Implementation of a Robust Ensemble Model: A novel ensemble approach is introduced for detecting cyberattacks in the ecosystem of the Internet of Medical Things (IoMT). Support Vector Machine (SVM) and Extreme Gradient Boosting (XGBoost) help to enhance the classification accuracy of the model and combat against difficult attack situations like man-in-the-middle (MITM) attacks.

1. **Enhanced Detection Accuracy and Practical Feasibility:** High classification accuracy of 98.04% is attained with WUSTL-EHMS-2020 dataset. This significant finding demonstrates the model's applicability by demonstrating its ability to accurately predict cyberattacks in the IoMT environment.
2. **Collaborative Employment of ML Algorithms:** The ensemble design utilizes XGBoost's proficiency in handling complex feature interactions and optimizing gradient-based learning, as well as SVM's capacity to accurately define decision boundaries in high-dimensional spaces. This synergy not only enhances the model's classification performance but also increases its adaptability to the distinct data properties of IoMT, ensuring reliability against a range of attack scenarios.
3. **Improved Cybersecurity Solutions for IoMT:** In response to the pressing need for effective cyberattack detection, a robust and reliable solution was introduced to the IoMT cybersecurity space.

The proposed approach ensures the confidentiality and security of patient data and healthcare operations by providing cybersecurity specialists and healthcare practitioners with a powerful tool to protect IoMT environments. This work encourages safer technology integration in healthcare by setting the base for future studies and advancements in secure IoMT systems.

The rest of the paper is organized with the following sections: Section II describes recently completed research works in detailed manner, section III details the needs of IoMT ecosystem and utilized datasets, section IV provides comprehensive view of the proposed system its important components, section V discusses and compares the results of proposed system and finally the conclusion is provided for this novel research work.

Related work

The rapid expansion of interconnected medical devices has intensified research interest in IoMT security⁶, as heterogeneous device behavior and large-scale connectivity expose healthcare networks to complex attack surfaces⁷. Here, recently proposed IoMT based intrusion detection frameworks are detailed. Summary of the recent research works are provided in Table 1. IoT driven eHealth system is evaluated qualitatively on the basis of theoretical and practical aspects⁸. Issues related to knowledge management, business plan, implementation, security and privacy are discussed in detailed manner.

Usman et al.⁹ introduced a privacy-preserving IoMT data aggregation framework that employs clustered multimedia sensor networks and cloud-based semantic analysis, demonstrating improved detection reliability through controlled experimental evaluation.

An early anomaly-driven intrusion detection framework tailored for IoMT environments employed mobile agents and machine learning techniques to jointly analyze network-level and device-level behaviors¹⁰. Autonomous mobile agents are used to integrate medical devices with attack detection mechanism. Sensor agent (SA), cluster-head agent (CH) and detective agent (DA) are three kinds of active agents. Machine learning algorithms and regression algorithms are employed for detecting network intrusions as well as data anomalies. Each and every associated ML algorithm is evaluated for selecting appropriate intrusion detection algorithm at network level as well as device level.

Similar AIDS is proposed for protecting IoMT edge network and the associated devices and gateway of IoMT from security vulnerabilities and exploitations related to IoT¹¹. It is contained with a collection of components of distributed monitoring and data acquisition (MDA). Log files which are collected from devices and gateway of IoMT and network traffic of IoMT are considered for computation cost. ML techniques are applied for detecting malicious and abnormal incidents in IoMT edge network. TON_IoT Telemetry dataset's network part is utilized and it is incorporated for training and testing with the familiar ML algorithms like DT, NB, LR, RF, SVM, and KNN. Based on the results DT, RF, and KNN algorithms provide better performance in detection.

An examination of recent IIoT and IoMT security frameworks reveals key design trends, particularly in balancing detection accuracy, computational cost, and deployment feasibility. Elwahsh et al. implemented a meta-heuristic optimization framework with hyperparameter tuning in automated manner by applying altered Harris Hawks Optimization to CNN-SVM and XGBoost for improving accuracy, robustness and model stability for resolving the limitations of orthodox IDS models¹². Likewise, the research works of Tian et al. focused on adversarial attack research especially EVADE and FDI frameworks expose the intelligent attacker perspectives in evading and breaking the security systems and highlights the importance of adversarial resilience in protecting IoMT rather than classification accuracy¹³.

Further considering the lightweight and resource constraints in IDS designs, improved GA-driven LSTM frameworks and GA-mADAM-IoT¹⁴ used attention mechanisms, genetic optimisation and effective feature selection by optimizing temporal-based models of edge environments to accomplish high accuracy with minimal latency, computation and memory¹⁵. In addition, works related to explainable IDS and privacy-preserving systems like CPS-IIoT-P2Attention and XAIEnsembleTL-IoV integrated explainable transfer-learning with scaled dot-product attention to assist zero-day attack tolerability in healthcare systems^{16,17}. Multi view models are demonstrated in some of the hybrid ensemble architecture of SCADA and deep composite model combined DCNN, autoencoders and LSTM which provide protection against anomalies over critical infrastructure systems by capturing the complexities of spatial-temporal telemetry¹⁸. These recent works provide clear insights for designing IoMT IDS and securing IoMT environments¹⁹.

Swarm-Neural Network (Swarm-NN)²⁰ approach is proposed for detecting attacks in IoMT edge network making use of Empirical Intelligent Agent (EIA)²¹. It is aimed to discover and analyse attacks during network communication of health care data in IoMT edge network. Proposed method is assessed by utilizing the real time protected ToN-IoT dataset. Experimental validation on the ToN-IoT dataset indicates that the Swarm-NN approach achieves near-perfect classification performance with minimal misclassification. Further proposed method is compared with classical ML models such as SVM, LR, DT and KNN.

Reference study	Input features / representation	Learning technique	Detection paradigm	Dataset(s) used	Learning nature	Target domain	Performance indicators	Core contributions
Nasayreh et al. ⁶²	LSTM- + PCA	KNN	Anomaly-based	WUSTL-EHMS, TON-IoT, ICU	Hybrid	IoMT Healthcare Systems	Accuracy, Precision, Recall, F1	Strong robustness against diverse cyberattacks
Ghourabi & Alkhali ⁵⁷	Federated feature learning across clients	XGBoost (Federated)	Anomaly-based	WUSTL-EHMS, CICIoMT	Hybrid	Distributed IoMT Networks	Accuracy, Precision, Recall, F1	Enables privacy-preserving intrusion detection
Zhang et al. ⁵⁴	Feature-optimized IoMT traffic features	ML-based IDS (feature-optimized)	Anomaly-based	ICU, WUSTL-EHMS-2020	Anomaly	IoMT Healthcare Networks	Accuracy	Significantly improves IDS accuracy
Zukaib et al. ⁵⁶	Meta-learning features from distributed IoMT nodes	DT, RF, AdaBoost, Extra Trees + XGBoost (meta-learner)	Hybrid (Known + Zero-day)	M-IoT-Env	Hybrid	Fog-Cloud IoMT Networks	Accuracy (near-perfect)	Meta-learning enhances generalization
Dhanya & Chitra ⁵⁴	IoMT device telemetry	XGBoost	ML-based anomaly detection	IoMT	Anomaly-based	IoMT Environments	Accuracy	Effectively for abnormal behavior detection
Rani et al. ³³	Service-level IoMT application data	ML-based anomaly detection	Anomaly-driven	IoMT	Anomaly-based	IoMT Services	Accuracy, Detection Rate	strong service detection capability
Radoglou-Grammatikis et al. ³¹	EC60870-5–104 protocol features and threat metrics	CART	SDN-enabled IDPS	N/A	SDN-based	Industrial Healthcare Systems	Accuracy, Adaptability, Detection Rate	adaptive defense
Izhar et al. ³⁰	Explainable feature representations (XAI)	XSRU	DL-driven IoMT detection	TON-IoT	DL-driven	IoMT Networks	Accuracy, Explainability	deep learning-based threat detection
Haseeb et al. ³²	Network traffic statistics	ML-SDN-based anomaly detection	Anomaly-oriented	N/A	Anomaly	IoMT Networks	Accuracy, Precision, Recall	operational efficiency in SDN-IoMT setups
Soneila & Adnan ²⁹	Malware behavior with cross-validation	CNN-LSTM	Hybrid DL detection	IoT Malware	Hybrid DL	IoMT Malware Analysis	Accuracy, Precision, Recall	intelligent detection of IoMT malware variants
Prabhat et al. ²⁶	Fog-cloud IoMT traffic features	DT, NB, RF	Ensemble-based IDS	TON-IoT	Ensemble	IoMT Networks	Accuracy, Precision, Recall	robust cyber-attack detection
Saheed et al. ²³	PSO-optimized NSL-KDD features	SVM, RF, DT, NB, RNN	DRNN-SML framework	NSL-KDD	SML-based	Smart IoMT Environments	Accuracy, Detection Rate, Training Time	reduced training overhead
Radoglou-Grammatikis et al. ²²	Modbus/TCP and HTTP traffic	DT, RF	Active-learning IDPS	N/A	Active learning	Healthcare Systems	Detection Rate, Adaptability	continuous self-learning
Nandy et al. ²¹	Intelligent agent-based features	Swarm-NN	Hybridized IDS	TON-IoT	Hybrid	Secure IoMT Networks	Accuracy, Precision, Recall	reliable intrusion detection
Zachos et al. ¹¹	TON-IoT telemetry attributes	DT, NB, LR, RF, SVM, KNN	Anomaly-based IDS	TON-IoT	Anomaly-based	IoMT Networks	Accuracy, F1-Score	strong anomaly detection
Thamilarasu et al. ¹⁰	Hierarchical autonomous features	ML & regression models	Mobile-agent IDS	N/A	Anomaly-based	IoMT Systems	Detection Rate, False Positives	Reduces false alarms
Shahzana et al. ²⁸	CNN & cuDNN-LSTM features	CNN, cuDNN-LSTM	SDN-enabled hybrid DL	Bot-IoT	Hybrid DL	IoMT Networks	Accuracy, Detection Time	Effective for evolving IoMT cyber threats
Swarna Priya et al. ²⁷	One-hot encoded attributes	KNN, NB, RF, SVM, DNN	PCA-GWO-based DNN	Kaggle	Hybrid	IoMT Architecture	Accuracy, Precision, Recall	advanced feature engineering
T. Saba ²⁵	KDDCup'99 features	Bagging, Boosting, Voting	PCA-ensemble IDS	KDDCup'99	Ensemble	Smart Healthcare Facilities	Accuracy, F1-Score	Improves IDS reliability
Continued								

Reference study	Input features / representation	Learning technique	Detection paradigm	Dataset(s) used	Learning nature	Target domain	Performance indicators	Core contributions
Manimurugan et al. ²⁴	Multi-attack traffic patterns	DBN	DBN-IDS	CICIDS-2017	Anomaly-based	IoMT Security	Accuracy, Detection Rate	high detection rates
Usman et al. ⁹	Sensor-level cluster features	CP-ANN	P2DCA framework	N/A	Privacy-preserving	IoMT Applications	Accuracy, Latency	data privacy and detection efficiency

Table 1. Summary of recent works.

Intrusion Detection and Prevention System is implemented for healthcare systems which utilizes the services of Modbus/TCP and HTTP protocols²². Active learning methodology is applied for detecting intrusions in which supervised classifiers are retrained dynamically. While considering HTTP protocol, attacks which are identified and diminished by the proposed IDPS are Denial of Service (DoS), SQL Injection (SQLi), Brute force attack and Cross-Site Scripting (XSS). Similarly, for Modbus/TCP protocol, 14 Modbus/TCP oriented attacks are detected. Based on the performance analysis of the propose IDPS, Decision Tree attained first position for HTTP protocol and Random forest achieved top position for Modbus/TCP protocol.

Deep Recurrent Neural Network (DRNN) and supervised Machine Learning (SML) based classification model is proposed and utilized for detecting IoMT ecosystem related intrusions making use of NSLKDD datasets²³. It focused on IoMT ecosystem related attacks such as DoS, probing, u2R and local attacks. By using this technique smart medical appliances communicate peer to peer with distinct IP addresses. Particle Swarm Optimization (PSO) method is incorporated for pre-processing and feature selection phases of proposed model. Further classification is performed using state-of-the-art machine learning algorithms like SVM, RF, DT, RC, NB, and RNN. Results depict that 99.76% accuracy is attained by this model.

DBN (Deep Belief Network) based intrusion detection system is proposed. DBN-IDS used CICIDS 2017 dataset for improving the performance of IDS in order to detect attacks and anomalies²⁴. CICIDS is a labelled dataset which provides various kinds of attacks like DoS/DDoS, Botnet, Brute force, Web attack, Infiltration and portscan. Performance of the DBN-IDS is evaluated based on the metrics such as detection accuracy, recall, precision and F1 score. Detection accuracy of DBN-IDS is attained as 99.37%, 97.93%, 97.71%, 96.67%, 96.37%, 97.71% and 98.37% for normal class, Botnet class, Brute Force class, Dos/DDoS class, Infiltration class, Ports can class and Web attack class respectively. To ensure secure smart city hospitals environment, IDS is proposed by utilizing Principal Component Analysis (PCA) and ensemble based classifiers²⁵. Feature reduction is performed by PCA and ensemble based classifiers are utilized for predicting intrusion attacks of the environment. KDDCup-'99' dataset is used and result displays that 93.2% accuracy is attained by the proposed IDS.

IoMT cyber-attack detection framework is proposed by utilizing ensemble learning and fog cloud architecture²⁶. It is distributed security system which has two primary engines which are traffic processing engine and intrusion detection engine. In traffic processing engine, correlation coefficient method is used for feature selection and min max method is used for feature normalization. Final detection model is designed making use of XGBoost based ensemble method which merges DT, NB and RF. Intrusion detection engine utilizes ToN-IoT dataset for evaluation purpose with valid evaluation metrics. Based on the results it attained 99.98% detection rate, 96.35% accuracy and it has the ability to shrink false alarm rate till 5.59%.

Hybrid PCA-GWO based DNN Classifier model is proposed to classify the intrusion attacks making use of Kaggle data set²⁷. Feature reduction and feature extraction are performed by DNN model. One-Hot encoding scheme is applied for pre-processing the core IoMT dataset. First level reduction is done at the pre-processed dataset making use of PCA. Grey-wolf optimization method is applied at the second level. After that reduced dataset is forwarded for classification by employing familiar state of art ML algorithms KNN, NB, RF, SVM and DNNs. Based on the results it is observed that accuracy is increased by 15% and time complexity is reduced by 32%.

Hybrid DL-driven SDN-enabled IoMT detection framework is proposed for combating against crucial attacks such as DDoS, data theft and reconnaissance²⁸. To detect multivector malware attacks effectively, Convolutional Neural Network(CNN) and Cuda Deep Neural Network Long Short Term Memory (cuDNNLSTM) are utilized. State of the art IoMT dataset namely Bot-IoT dataset is employed for evaluation with primary performance metrics. Attained results of Hybrid DL-driven SDN-enabled IoMT detection framework are 99.99% accuracy, 99.83% precision, 99.33% recall and 99.33% F1 score.

Similarly, another hybrid DL-driven SDN-enabled framework (Hybrid CNN-LSTM) is proposed in order to detect classy IoMT malwares²⁹. It integrates CNN and LSTM IoT Malware Data set is utilized for complete evaluation process. tenfold cross validation is employed for depicting results as entirely unbiased. Performance is evaluated with the help of standard as well as extended metrics. Detection accuracy is attained as 99.83% with 1.2 s time complexity on the basis of experimental result.

XSRU-IoMT model is proposed for detecting sophisticated attack vectors of IoMT system³⁰. It is implemented making use of bidirectional recurrent units (SRU) for fastening training process of recurrent networks. Explainable AI (XAI) ideology is adopted for increasing trust level where predictive decisions related explanations are provided for ensuring secure environment. Testing is performed using ToN-IoT dataset. This SRU driven DL model has the ability to ease vanish gradient problem and detect various cyber threats and attacks of IoMT system such as DoS, injection and data breaching attacks. The evaluation confirmed strong detection

performance across standard classification metrics, highlighting the model's robustness against multiple IoMT attack categories.

IDPS is proposed by integrating ML and SDN (Software Defined Network) for detecting and mitigating cyber-attacks of EC60870-5-104³¹. EC60870-5-104 protocol is utilized by most of the industries of the healthcare system. Quantitative threat model is presented to assess the cruelty of cyberattacks related to EC60870-5-104. Intrusion detection is based on CART classifier which utilizes the flow statistic of both TCP/IP network and EC60870-5-104 pay load. Further, SDN based mitigation method is applied in which automatic mitigation is converted as MAB problem and solution is provided by TS and SDN. On the basis of experiment results, 0.831 detection accuracy and 0.8258 F1 score are obtained with 0.923 mitigation accuracy.

Haseeb et al. addressed disputes in the health sector of application within the smart cities using IoMT, focusing on efficient data and in the network domain of security³². It encompasses a machine learning model with SDN-enabled securial system to anticipate the network resource and encoorp the sensor data. By the unsupervised machine learning and dynamic prediction of metric, it decreases the communication and improves the routing performance. The Software-Defined Networking model reduces the Krizhevsky control plane overhead and combines with the routing scheme to configure a secure network. Early Outcomes show the important improvements in network through the output, data dropping, and data delay. The Future work of this paper aims to provide the scalability and flexibility with various user controllers and to develop proper accuracy using deep learning methodology.

Rani et al. enhances the SmartHealth, a Machine Learning model designed to improve security in the Smart Healthcare Systems – SHS³³. This SmartHealth model visionize the signs from the IoMT devices, as they concurrete the difference between security attacks and normal activities. An accuracy of 92% is being achieved for the identification of the unaware activites with a F1-score of 90% achievement by the norms of performance analysis. This framework defines for identifying many attacks to address a broad way of security threat detection. The ML model offers promising the potential for scaling up a huge range of security threats in the upcoming days as the consequence is to drench the IOMT devices in a secure way and safe guard patient data. Dhanya and Chitra putsforward an intelligent classy framework for providing security for the Internet of Medical Things (IoMT) data through an Autoencoder dimensionality approach collaborated with the XGBoost Classifier proposed with Genetic Algorithm³⁴. It optimizes resource usage and useful in saving the time, and is implemented on the fog layer which offers a light solution for malware detection without proper feature selection. The approach is not dependent on the dataset features as it presents a method for incrementing proper accuracy in IoMT devices.

Recent studies indicate that data-driven learning paradigms are increasingly central to IoMT security, as rule-based defenses fail to cope with evolving and device-specific attack behaviors. Although ensemble learning approaches, combining boosting algorithms and support vector machines, have been explored in generic IoT intrusion detection systems, direct applications of such techniques in IoMT environments remain limited because of their own unique operational and safety-critical nature of healthcare systems³⁵. IoMT deployments integrate heterogeneous medical sensors, physiological data streams, and real-time monitoring requirements, where false negatives can directly threaten patient safety. The proposed model extends the state-of-the-art by fusing XGBoost with SVM classifiers using both RBF and polynomial kernels for complementary learning from the combined network-flow and biometric features provided by WUSTL-EHMS-2020 dataset. XGBoost learns complex nonlinear interactions between features and conducts embedded feature prioritization. SVM-RBF effectively detects localized anomalies, while the SVM-Polynomial models higher-order global decision boundaries. This tri-model ensemble enhances robustness against spoofing and data injection attacks particularly relevant to healthcare monitoring systems³⁶.

Preliminaries

This section provides appropriate information regarding IoMT ecosystem, Algorithms used and datasets which are utilized by the proposed IDS training and validating the realtime performance.

IoMT ecosystem

Nowadays, Internet-of-Things (IoT) usage has been increased in our day-to-day life with various applications. As per the report of Garner, it is predicted that before the end of the year 2020, the amount of IoT devices have been increased by 20% when compared to the previous years. In 2021, the IoT active devices worldwide are increased up to 26.66 billion because of their compactness, different ways of communication and low consumption power during the data processing.

Internet of Medical Things (IoMT) is one of the branches of IoT used in the health industry for a purpose of monitoring a patient to diagenesis the disease. IoMT is developed by the combination of both hardware and software with physical connections in between the internet with the help of wireless technologies. IoMT is connected to various sensors to collect continuous information, wearable devices to monitor human activities, medical devices, and also clinical systems³⁰. Wearable devices are also most popular in health care industries and academic purposes in recent days. The Importance of wearable devices and the IoT in the market gives an actual growth of these technologies. IoMT is useful for both the doctors and patients, while the implementation is a challengeable task for the vendors/organizations.

The source from Rick martin stated the quote “Like most technology, IoMT brings pros and cons” which are discussed along with the challenges. There are three major challenges to implementing an IoMT. First one is deploying a device on healthcare is a higher cost in terms of replacing the last technology and to develop an IT infrastructure network to communicate the data. Second one is to protect a data in hospital is a critical one because hospitals have hundreds of millions of data are saved in a cloud via the internet, all these captured data from a patient will always run a test on the IoMT to check the vulnerability in a dataset. Third one is installation

of IoMT is not a simple task whereby the devices connect it to hospital network is a harder one to prevent the data in a secure manner.

IoMT eco system is comprised with the four primary components which are Perception layer, Connectivity layer, Processing layer and Application layer. In perception layer, all types of health related data are collected making use of various smart medical devices. Connectivity layer implements Bi directional data transmission amid perception layer and cloud (such as AWS, MS Azure, Google Cloud) by utilizing networks and gateways. Processing layer acts as a cloud middleware for data storage and management. Application layer allows end users for analysing data, reporting and provide device integrated app management. Typical IoMT ecosystem is depicted in Fig. 2 where vast e-health care gadgets are interconnected with physicians for providing effective aid and treatment. Cloud hosted medical data is dispensed to users through distinct gateway.

Based on Cisco, a normal hospital room is contained with minimum 15 to 20 connected devices which have the possibility of 6.2 cyber security exploitations amid them as an average. Malicious hackers and cyber criminals are having eagerness in targeting them to obtain medical records as well as PII (Personally Identifiable Information) of patients. Like IoT, IoMT also having various general vulnerabilities. After observing those vulnerabilities and their impacts only it is possible to secure the medical devices and system from attacks.

All the medical devices are provided with advanced wireless technology like Bluetooth, WIFI, etc. that may allow to communicate the data from machine-to-machine which is a basic structure of IoMT environment. Security is a major problem for IoT environment because all the nodes are placed in dynamic manner. With this impact, the attackers are possible to breach the information from the nodes while communicating a data via internet. This becomes a major issue subject to various types of attacks in IoMT that interact with communication and to control the smart devices. Traditional methods are not efficient cyber attack detection in a IoT/IoMT. Mirai and Bricker Botnets are a recent attack to breach the data.

IoMT is a system of medical devices connected through internet. It integrates hardware, software and communication medium for connecting healthcare IT. IoMT is applied in various forms like sleep monitoring, heart monitoring, blood glucose monitoring, remote temperature monitoring, and remote biometric scanning and so on. Weak authentication process, accessibility of IoMT devices from external devices, unpatched software, insecure network access and device loss are considered as the general vulnerabilities of IoMT.

To ensure strong authentication process multi factor authentication schemes and strong authentication protocols should be utilized. Device accessibility of IoMT require only authorized personnel with strong password schemes. It is mandatory to ensure quick installation patches without any exploitation intervention. IoMT network segmentation will reduce the insecure network access of intruders. If any IoMT device is lost they can be saved from attackers only by incorporating strong authentication and tracking methodologies.

Algorithms used

SVM(Poly, Rbf)

The Support Vector Machine (SVM) identifies the optimal hyperplane in a high-dimensional feature space to effectively separate different classes of data points. Categorizing malicious and benign actions is an essential role of SVM in safeguarding the Internet of Medical Things (IoMT) from sophisticated attacks. These categories are established based on certain attributes derived from system logs, network traffic patterns, and device activities.

The Radial Basis Function (RBF) kernel and the Polynomial (poly) kernel are two prevalent kernel functions in SVM that enhance its ability to handle diverse data dispersion and complex decision boundaries. The RBF kernel excels at modeling non-linear interactions in high-dimensional environments, while the poly kernel effectively captures polynomial correlations. IoMT security systems can successfully identify and classify cyber threats, including ransomware, probing, and denial-of-service (DoS/DDoS) assaults, utilizing both kernel types.

The nature of the data and the equilibrium between model difficulty and generalization dictate the optimal choice between the polynomial and radial basis function kernels. Employing SVM with these kernels alongside other advanced machine learning methodologies, such as Extreme Gradient Boosting (XGBoost), is crucial for ensemble models, enhancing accuracy and reliability to safeguard IoMT ecosystems.

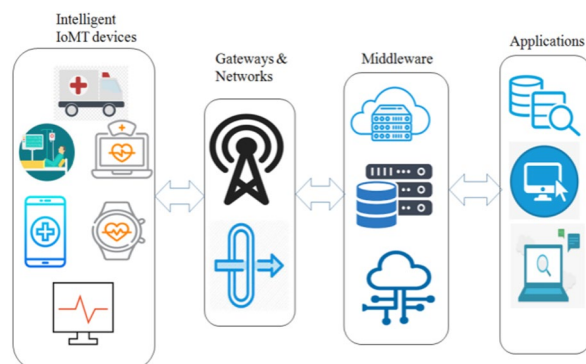


Fig. 2. IoMT Ecosystem.

XGBoost

An advanced machine learning technique known as XGBoost (eXtreme Gradient Boosting) is crucial for attack categorization within the Internet of Medical Things (IoMT) ecosystem. XGBoost employs ensemble learning techniques to create a succession of decision trees, with each tree systematically correcting the mistakes of its predecessor. Then, XGBoost employs an iterative refining technique to generate a model that is highly accurate and resilient, effectively distinguishing between harmful and benign activities in IoMT contexts.

The ability in managing instable datasets is considered as a primary feature of XGBoost. XGBoost ensures equitable illustration of both minority and majority classes through approaches such as weighted learning and boosting, improving the model's skill to perceive rare yet significant cyber risks. XGBoost is ideally suitable to manage large and complex datasets prevalent in IoMT systems due to its scalability and efficiency. XGBoost provides a comprehensive approach to safeguard IoMT devices from intricate and continuously growing threats by improving the efficacy and robustness of ensemble models when integrated with supplementary algorithms such as Support Vector Machine (SVM).

Hybrid ensemble approach: SVM and XGBoost integration

Incorporating XGBoost into the ensemble boosts the model's capability to identify composite patterns in cyberattacks specific to IoMTs and adapt to the dynamic, complex behaviors of threats within medical device ecosystems. The classification approach leverages the complementary strengths of both algorithms by merging SVM and XGBoost into a hybrid ensemble. XGBoost enhances the model's robustness and scalability through gradient-based optimization and effective management of imbalanced datasets, whilst SVM excels in delineating precise decision boundaries essential for detecting subtle assault patterns. This ensemble technique effectively addresses the exclusive security challenges of IoMT, such as resource limitations, the private nature of health data, and the diverse connection of devices. This method enhances health care data security and ensures the functioning integrity of interconnected healthcare systems by providing a reliable and scalable cyber attack detection system.

Datasets

WUSTL-EHMS-2020, CICIDS 2017, TON-IoT, KDDCup-'99' and Kaggle are several IoMT datasets used for experimentation in recent state of the art works³⁷. WUSTL-EHMS-2020 dataset is mainly used by the proposed model for attaining better results in training as well as testing²⁴. Further, the effectiveness of the proposed system is validated using TON-IoT and CICIDS 2017 datasets.

WUSTL-EHMS-2020: EHMS (Enhanced Healthcare Monitoring System) testbed created this realtime dataset³⁸. It is contained with metrics related to both network flow and biometrics. EHMS testbed is classified into 4 divisions which are gateway, sensor, network and switch and router visualization. ARGUS (Audit Record Generation and Utilization System) tool is used for capturing and storing network traffic and biometrics data as CSV file. Size of dataset is 4.4 mb which includes 16,318 samples totally among them 87.5% samples are normal samples and remaining 12.5% samples are attack samples. The dataset contains:

Attack data

Attack data refers to datasets containing information about cyber attacks like DoS and Data injection and their characteristics. For cybersecurity researchers, practitioners, and analysts looking to understand, assess, and counter numerous IoMT threats, these datasets are precious resources. Attack data often contains a widespread range of characteristics and pieces that define the behavior, characteristics, and operation of different attack situations. Malware detection, vulnerability assessment, threat intelligence analysis, incident response, and security solutions development such as antivirus software, threat detection algorithms and intrusion detection systems are just a limited of the tasks it performs in cybersecurity research and protection. Cybersecurity professionals may be able to identify patterns and trends in attack activity and better comprehend new risks by using this information.

Benign (no attack) data

In the given dataset, benign data refers to instances that demonstrate acceptable or benign behavior within the IoMT ecosystem. This information describes the typical behavior of IoMT devices operating under normal conditions, devoid of any malicious intent or activity. Numerous actions, exchanges, and transactions carried out by IoMT devices in healthcare settings, including as patient monitoring, data transmission, device connectivity, and system maintenance, are included in the category of benign data. Cyber security analysts can develop efficient detection systems and anomaly detection techniques to strengthen IoMT systems against security threats and preserve the integrity and privacy of health care data by analyzing standard data and identifying deviations or abnormalities³⁹. Utilizing the perceptions of standard data facilitates the employment of proactive safety procedures to combat against cybersecurity threats and exposures.

Proposed methodology

Cyberattacks of IoMT ecosystem are detected by adapting state-of-the-art machine learning techniques. Here, the proposed methodology is thoroughly described, including the methods and algorithms used to identify and reduce the risks of cyberattacks and vulnerabilities. Figs. 3, 4 shows the architecture of hybrid ensemble model and Fig. 3 affords the complete process of proposed method. The suggested ensemble model uses the advantages of both algorithms—Support Vector Machine (SVM) and Extreme Gradient Boosting (XGBoost)—to provide reliable and accurate cyber threat detection in IoMT devices.

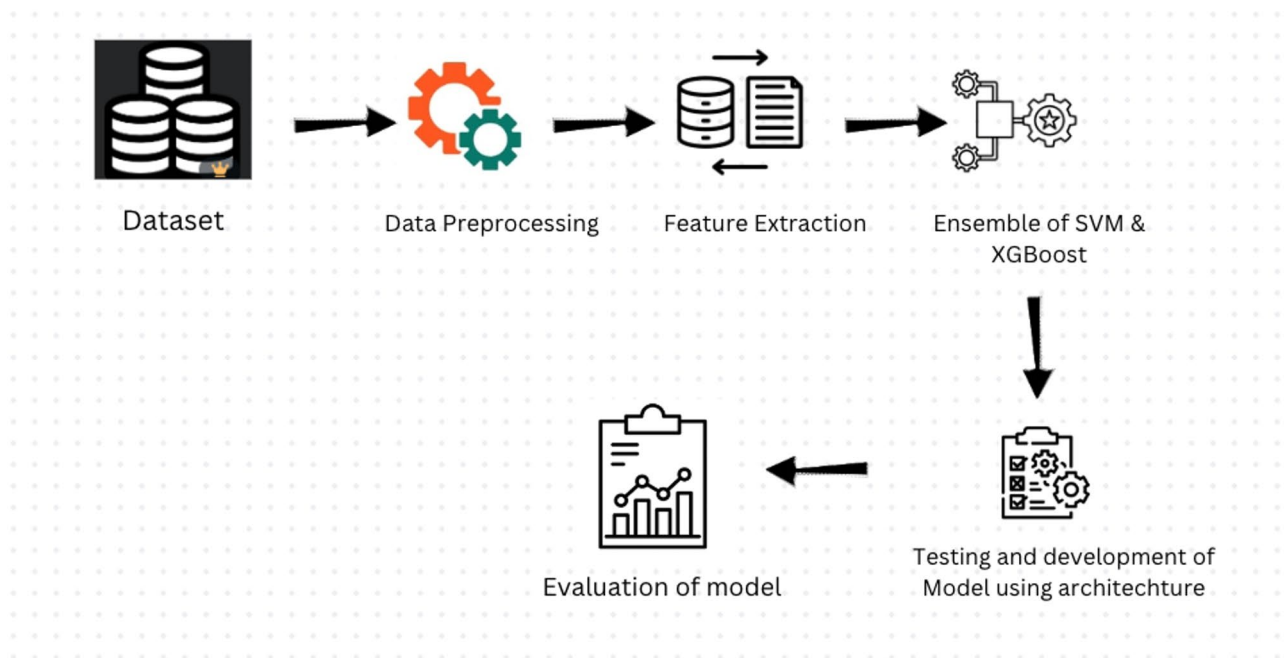


Fig. 3. Work Flow of proposed model.

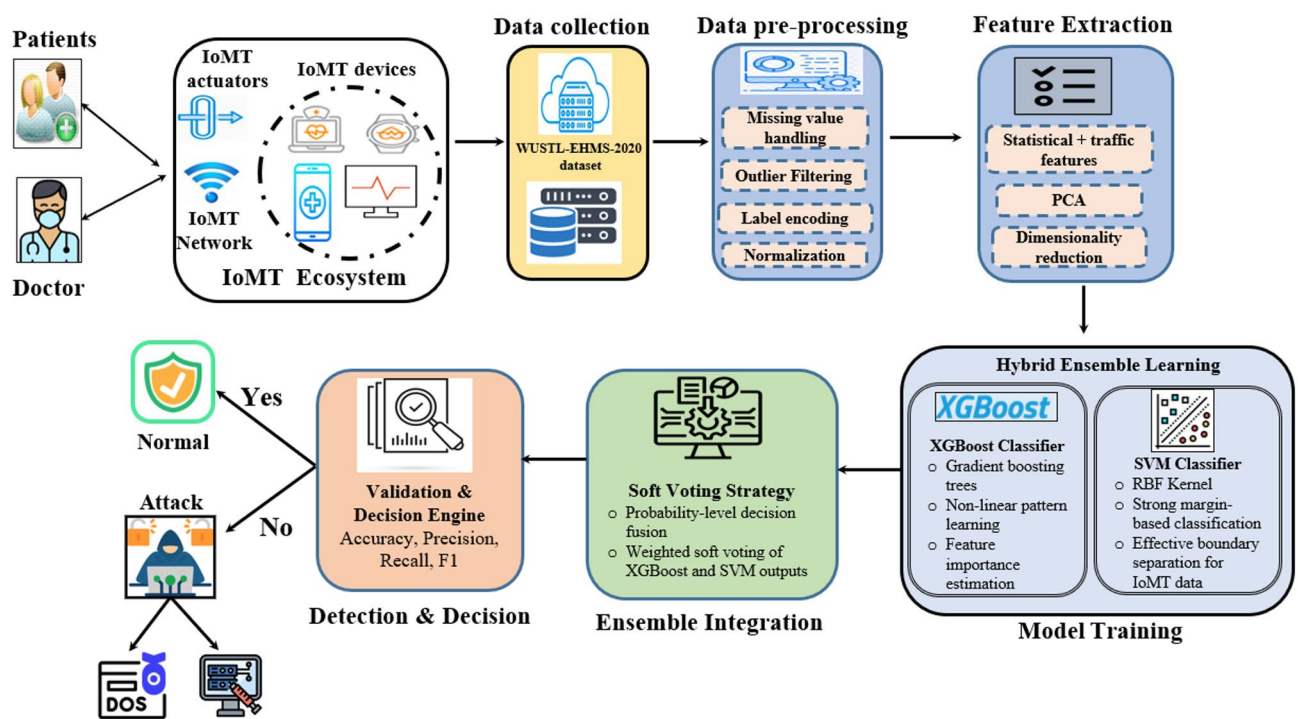


Fig. 4. Proposed Methodology.

Proposed system

As IoMT ecosystem is a internet connected network of multiple devices and sensors, health status of patients is examined and quick periodic updates are sent physicians to maintain healthy ecosystem. Delicate data collection is performed using those smart devices and collected data is stored in servers. Though IoMT devices are intelligent they are not smart enough regarding the safe delivery of data and intrusion detection. Multiple attack vectors are targeted by the attacker/intruders to impair the system and end-users by infecting with the attacks like DoS/DDoS, MIM, probe attack, impersonation attack and so on.

The objective is to develop a robust machine learning-based system capable of accurately distinguishing between malicious (attack) and non-malicious (normal) activities within IoMT devices. SVM and XGBoost algorithms are leveraged for analysing several features which are extracted from IoMT devices to improve accurate classification against cyberattacks. The main goal of this research work is to design an enhanced cyber attack protection model for IoMT environment by securely handling patient data and combating against significant cyber threats of IoMT network. In addition, it aims to implement realistic applicable system for smart healthcare by overcoming the limitations related to IoMT devices and performance by conducting experimentation and validation. By implementing task allocation, model performance is optimized to show the efficacy of the model. To detect the attack patterns of IoMT ecosystem effectively, model involved rigorous noise identification and removal which is considered as essential to detect and mitigate attack threats. Classification accuracy is improved by conducting the complete testbed including training, validation and hyperparameter tuning by employing SVM and XGBoost models to assure the robust performance.

Data collection

Data collection phase involves accumulating data from various components of IoMT ecosystem such as IoMT gateway, Network and IoMT devices like context aware sensors, actuators and bio sensors. Gateway and servers act as intermediaries to forward the sensitive data of sensors and devices and data can be eavesdropped by the malicious people to impair the security standard. WUSTL-EHMS-2020 dataset is used as primary for this research work.

Feature selection

Dataset quality is enhanced by eliminating the unsolicited data in this data preprocessing phase. To select the optimal feature, this phase acts as first stage and the attained outcome is considered. Principal Component Analysis (PCA) is employed for optimizing selected features and the final outcome is forwarded further to ML based training and testing phase.

Proposed methodology

The proposed framework adopts the soft-voting ensemble strategy, rather than stacking or weighted voting. XGBoost, SVM with RBF kernel, and SVM with polynomial kernel are trained in parallel as three independent base learners. During inference, each classifier outputs class posterior probabilities, while the final decision is provided by computing the mean value of the outputs. The aggregated class probability values are computed, and the class label with the maximum accumulated probability is returned as the ultimate decision. To this purpose, such an architecture has been adopted, which is lightweight enough for IoMT gateways while exploiting the complementary decision boundaries brought by tree-based and margin-based classifiers with favorable time complexity. In that way, it outperforms deep neural networks which require high computation and energy demands.

Besides, the proposed ensemble is appropriate for resource-constrained IoMT environments, where computational efficiency, reliability, and generalization from limited labeled data are salient design requirements. Unlike deep learning-based intrusion detection systems that require very high-memory and computationally intensive processing architecture, the XGBoost-SVM (RBF-Poly) ensemble achieves a strong detection performance at a relatively moderate model complexity, thus it can easily be deployed at IoMT gateways or edge nodes. The margin-based learning by SVM reduces false negatives while XGBoost enhances the discrimination among heterogeneous device behaviors and noisy biometric signals. The complementary classifiers are combined using a soft-voting strategy; hence, the proposed framework strikes a good balance between accuracy, robustness, and computational cost, which sets it apart from other IoT-centric intrusion detection methods and further strengthens its viability for practical healthcare cyber-security applications³⁹.

Support vector machine

SVM is one of the machine learning techniques that is used for the binary classification used for classification of attack data in IoMT devices. SVM consists of three types of kernels in which these kernels are used to transform the data which will be easy to find the difference between the classes. The kernels are linear, poly, and rbf in which a linear kernel is not the correct one to use because it takes more time to execute the steps when compared to the remaining other two kernels which results in more time taken. So here we don't consider the linear kernel for the classifications that we have used the poly and rbf kernels. SVM divides data into two classifications for IoMT attack detection: normal and attack, based on the best hyperplane that divides the data^{40,41}. The formulation of the SVM optimization problem is:

$$\min_{w,b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \varepsilon_i$$

subject to:

$$y_i (\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 - \varepsilon_i, \varepsilon_i \geq 0,$$

where $\mathbf{w}$ is the weight vector, b is the bias term, ε_i are the slack variables for misclassified samples, C is the regularization parameter and y_i is the actual label (1 for attack, -1 for normal). The decision function for SVM is derived as follows

$$f(X) = \mathbf{w} \cdot \mathbf{x} + b$$

Polynomial kernel

The Polynomial kernel is one of the types in SVM in which it has a better understanding and it also explains how the model makes predictions when compared to other kernels. As the attack data consists of some of the complexities it analyses those complexities by not only considering the individual features but also considering the combinations of the data. By applying these models to the whustl dataset, we have obtained 94.21% accuracy.

RBF

It is another type of kernel in the SVM which also helps in identifying the complex patterns in the data to classify whether it is attack or not. The difference between the polynomial kernel and the rbf kernel is that the polynomial kernel undergoes the dot product between the given features and rbf measures the similarity between those features. The model attained an accuracy of 94.82%.

XGBoost & XGBoost pipeline

The XGBoost model exhibited outstanding performance, with an accuracy of 98.50% in the precise classification of assault patterns. Gradient boosting framework is used to separate the complex patterns of the IoMT dataset for achieving improved prediction accuracy. XGBClassifier enabled the model training with altered parameters for performance enhancement. This excellent performance highlights the XGBoost algorithm's effectiveness in handling complicated datasets and demonstrates its suitability for a variety of classification tasks⁴².

The XGBoost pipeline, with preprocessing methods such as target encoding, established notable efficiency in classifying attack patterns, accomplishing high accuracy. The pipeline's use of feature engineering approaches demonstrates its effectiveness in improving model performance, despite a little drop in accuracy when compared to the independent XGBoost model. Target encoding for the categorical feature transformation and an XGBoost classifier for the classification challenge are two of the pipeline's several stages. This all-encompassing approach ensures that the model uses enhanced feature representations in addition to learning from the data, resulting in robust predictions.

The objective function of XGBoost comprises a regularization term to regulate model complexity and a convex loss function to quantify the inconsistency amid the anticipated and real values. The objective function is delineated as follows:

$$\text{Obj} = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Here, L denotes the loss function (e.g., logistic loss for classification), y_i represents the actual label (1 for attack, 0 for normal), $\hat{y}_i$ indicates the anticipated likelihood that pattern i is attack and $\Omega(f_k)$ signifies the regularization term for the complexity of the model f_k .

The prediction for an instance i in XGBoost is derived as follows

$$\hat{y}_i = \sigma \sum_{k=1}^K f_k(x_i)$$

where σ is the sigmoid function, and f_k is the prediction from the k -th tree.

Ensemble learning

As we know the ensemble model gives better accuracy when compared to the individual model. Here we have combined the xgb model, xgb_pipeline, and svm_rbf by using the voting classifier which gives better accuracy when to the individual model. The soft voting Classifier is the ensemble technique. which combines multiple machine learning models to give better predictions. It undergoes different combinations of the machine models and gives the output at which we have the highest accuracy. It performs better than any other individual machine learning model. It also helps in reducing the overfitting and also increases the performance of the model. Combining these three models gives the accuracy of 94% which is the highest when compared to the individual machine learning model.

Instead of selecting the outcome of single classifier, the employed soft voting strategy process cumulatively generated posterior probability outputs of every model and weighted average is calculated. xgb model, xgb_pipeline, and svm_rbf are used as the base learners. The predicted probability of class c of sample x on k^{th} classifier is denoted by $P_k(c|x)$. Each classifier is associated with a weight w_k , where $\sum_{k=1}^3 w_k = 1$ $w_k \geq 0$ which ensures effective probabilistic fusion. The ensemble probability is computed as follows

$$P(c|x) = \sum_{k=1}^3 w_k P_k(c|x)$$

where $k \in \{\text{xgb}, \text{xgb_pipeline}, \text{and svm_rbf}\}$

Then, the final predicted class label is decided by choosing the class with high predicted probability. To ensure the detection reliability and improved classification accuracy, weighting factors w_k are decided on the basis of validation performance.

$$\text{Finalpredictedclass}(\hat{y}) = \arg \max_{c \in \{\text{Normal}, \text{Attack}\}} P_k(c|x)$$

Model	Hyperparameter	Value
XGBoost	Number of estimators	1000
	Maximum tree depth	10
	Learning rate	0.05
	Subsampling ratio	0.9
	Column sampling ratio	0.9
	Regularization parameter (λ)	1
SVM (RBF Kernel)	Kernel Type	RBF
	Regularization parameter (C)	1000
	Gamma	scale
	Probability enabled	True
	Random state	42
SVM (Polynomial Kernel)	Kernel Type	Polynomial
	Degree (d)	3
	Regularization parameter (C)	1000
	Gamma	scale
	Random state	42
Ensemble (Soft Voting)	Voting type	Soft
	Base estimators	XGBoost + SVM(RBF) + SVM(Poly)
	Weights	1,1

Table 2. Hyperparameter details.

Model	Inference latency (ms)	Model size (MB)	Throughput (samples/sec)	CPU utilization (%)	Memory usage (MB)
XGBoost	0.78	5.6	1282	14.2	62
XGB_Pipeline (PCA + XGBoost)	0.64	4.1	1562	12.8	55
SVM (RBF Kernel)	0.92	3.4	1086	16.5	58
Proposed Soft Voting Ensemble	1.14	9.2	877	22.1	98

Table 3. IoMT ease of deployment details.

In Table 2, explicit report of hyperparameter settings that allow the reproduction of the experiments has been added. In particular, the XGBoost classifier was set with 1000 estimators with a maximum tree depth of 10, learning rate of 0.05, subsampling ratio of 0.9, and column sampling ratio of 0.9. In turn, the RBF kernel was exploited in the SVMs by setting the gamma was set to "scale." The polynomial kernel exploited degree $d = 3$, while $C = 1000$ and the gamma was again set to "scale". According to these values, there is a balance amid detection accuracy and computational efficacy in resource-constrained IoMT environments. For addressing redundancy and noise in the high-dimensional feature space, Principal Component Analysis was carried out after feature encoding and normalization. This PCA configuration has been clearly stated in the revised methodology, together with its motivation in view of a lightweight deployment of the proposed approach in an IoMT environment. Table 3 depicts the details related to ease of deployment of the proposed work and Table 4 provides complete algorithm of the this ensemble model.

Results and comparative analysis

Evaluation metrics

After observing recent works and their limitations, performance of proposed system is measured by considering the metrics such as accuracy, recall, precision and F1 score⁴³. Before calculating performance metrics confusion matrix is computed as the initial step which is used for classifying instances of test dataset. True Positive (TP) represents properly categorized attack instance and True Negative (TN) represents usual instance⁴⁴. False Positive (FP) denotes improper classification of normal instance and False Negative (FN) represents improper classification of attack instances. Performance metrics used for evaluation are described as follows.

Accuracy

Accuracy reflects the proportion of correctly classified instances across both attack and normal classes in the evaluated IoMT datasets.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Algorithm: XGBoost–SVM (Soft Voting Ensemble) for IoMT Attack Detection

Input: IoMT dataset $D=\{D_{WUSTL-EHMS-2020},D_{ToN-IoT},D_{CICIDS-2017}\}$

Output: $y\in\{Normal,Attack\}$

- Dataset Preparation**
Split D into 70% training and 30% testing using stratified sampling.
- Preprocessing**
Handle lost values, encode unqualified features, and standardize numerical attributes.
- Dimensionality Reduction**
Apply PCA on training data and retain components preserving $\geq 95\%$ variance; transform both train and test sets.
- Train Base Models**
Train:
 - XGBoost on original features
 - XGBoost (Pipeline) on PCA features
 - SVM (RBF kernel) on original features
- Generate Probabilities**
Obtain class probability outputs from all three models for each test sample.
- Soft Voting Fusion**
Compute weighted average of probabilities
$$P_{ens} = w_1P_{XGB} + w_2P_{XGB_P} + w_3P_{SVM}$$
and assign class with maximum probability.
- Final Decision & Evaluation**
$$\hat{y}(x) = arg \max_{c \in \{Normal,Attack\}} P_{ens}(c|x)$$

Predict Attack / Normal and assess with Accuracy, Precision, Recall, and F1-score.

Table 4. Algorithm XGBoost–SVM (Soft Voting Ensemble) for IoMT Attack Detection.

Performance metrics	Value (%)
Accuracy	94.82
Precision	94.71
F1-Score	94.82
Recall	94.41
AUC-ROC	93.32

Table 5. Classification report for SVM (radial basis function).

Precision

It predicts the proposed syetem’s efficiency by computing the percentage of attack observation.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall

It depicts real time attack classification in terms of percentage for computing overall success rate of proposed IDS.

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

F1 score

Harmonious mean of precision and recall metrics are used for computing F1 score. To predict the accuracy it acts as statistical function.

$$\text{F1 score} = \frac{2 * TP}{2 * TP + FP + FN}$$

The classification report serve as crucial tools for assessing the performance of the SVM with radial basis function (RBF) kernel algorithm and the SVM with polynomial kernel in classifying data from Internet of Medical Things (IoMT) devices as either attack or normal. Classification reports are included with the complete values of accuracy, precision, recall, F1-score and AUC-ROC for both normal and attack classes which indicates the ability of classifying the instances precisely of both classes.

The classification reports provided in Tables 5, 6, 7 are considered as important assessment tools for the proposed ensemble model in term of classification accuracy against IoMT cyber-attacks. Table 8 shows the details of complete comparison of evaluation metrics with the testing datasets (WUSTL-EHMS-2020, Ton IoT and CICIDS-2017) used for this research work. Confusion matrices and ROC curves of respective validation datasets are illustrated in the Figs. 5, 6, 7, 8, 9, 10. This ensemble model is obtained by combing XGBoost, SVM

Performance metrics	Value (%)
Accuracy	94.21
Precision	94.18
F1-Score	94.21
Recall	93.60
AUC-ROC	90.95

Table 6. Classification report for SVM(polynomial).

Performance metrics	Value (%)
Accuracy	98.50
Precision	98.48
F1-Score	98.50
Recall	98.48
AUC-ROC	99.74

Table 7. Classification report of XGBoost.

Performance Metrics	Value for WUSTL-EHMS-2020 (%)	Value for Ton IoT (%)	Value for CICIDS-2017 (%)
Accuracy	98.04	99.26	99.84
Precision	98.03	99.52	99.38
F1-Score	98.04	99.54	99.69
Recall	97.99	99.53	99.53
AUC-ROC	99.65	99.87	99.99

Table 8. Classification report for ensemble of SVM (polynomial), SVM (radial basis function), XGBoost.

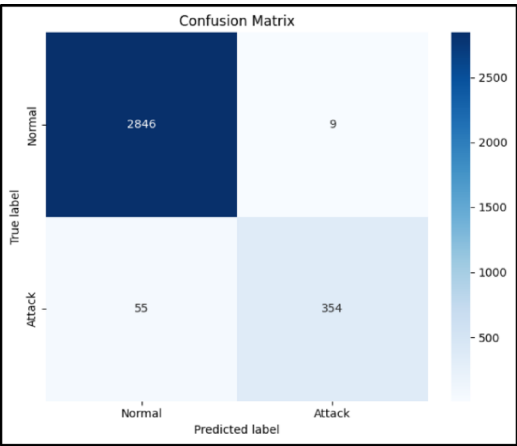


Fig. 5. Confusion matrix of the ensemble model (WUSTL-EHMS-2020).

with a polynomial kernel, and SVM with an RBF kernel and it outperforms individual models by attaining 98.04% accuracy.

Comparative analysis

A complete comparative analysis of the ensemble model is performed in this section. To validate the performance of the proposed model, accuracy, precision, F1 score, recall, AUC-ROC are used. Experimentation is conducted by considering WUSTL-EHMS-2020 dataset of Washington University in St. Louis as base for this study. The key potentials of XGBoost and SVM are amalgamated to develop the ensemble model. XGBoost was selected due to its ability to model complex feature interactions efficiently while maintaining scalability on structured IoMT

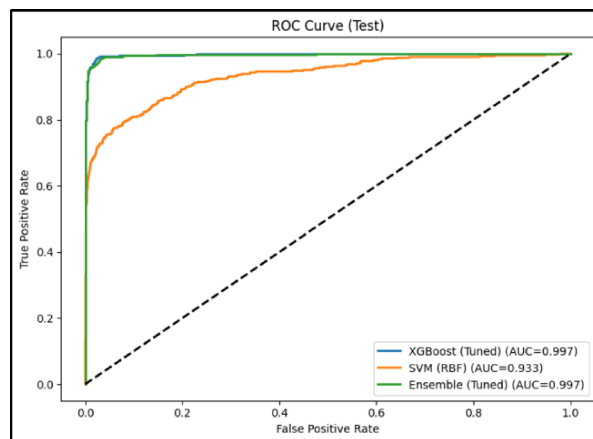


Fig. 6. ROC curve of the ensemble model (WUSTL-EHMS-2020).

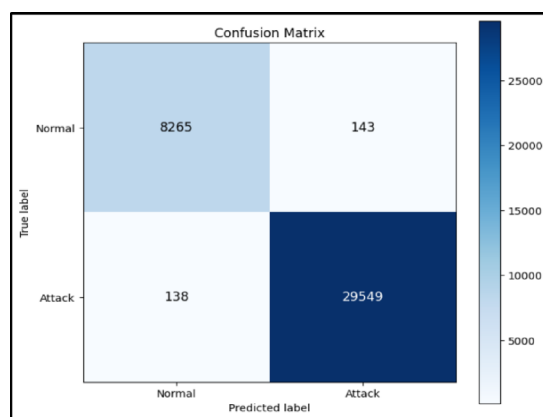


Fig. 7. Confusion matrix of the ensemble model (ToN-IoT).

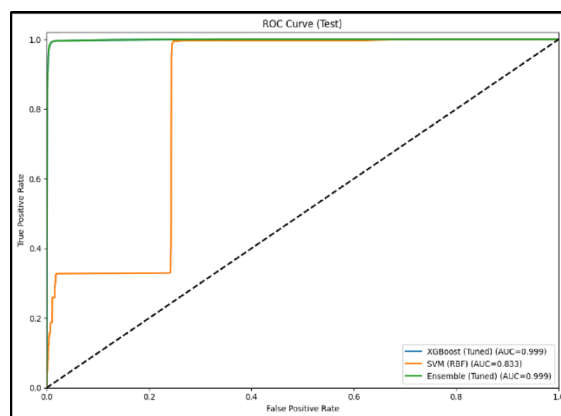


Fig. 8. ROC curve of the ensemble model (ToN-IoT).

traffic data. Likewise, SVM is highly effective in classification especially for high dimensional spaces. Across all evaluated metrics, the ensemble consistently outperformed individual classifiers, achieving over 98% accuracy on the WUSTL-EHMS-2020 dataset.

For ML based classification tasks, SVMs are considered with high priority. Rieck et al. employed SVMs for performing automated malicious behaviour analysis and attained 89.00% accuracy, 88.60% precision, 88.70% F1 score and 88.60% recall⁴⁵. Spatial hierarchies in data are particularly well-suited for Convolutional Neural Networks (CNNs). Hardy et al. developed DL4MD, a deep learning framework called which utilized CNNs

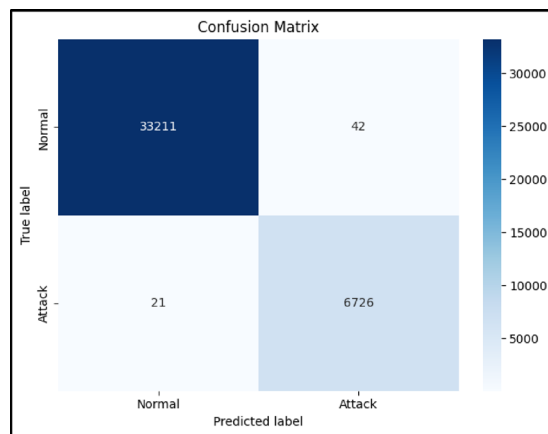


Fig. 9. Confusion matrix of the ensemble model (CICIDS-2017).

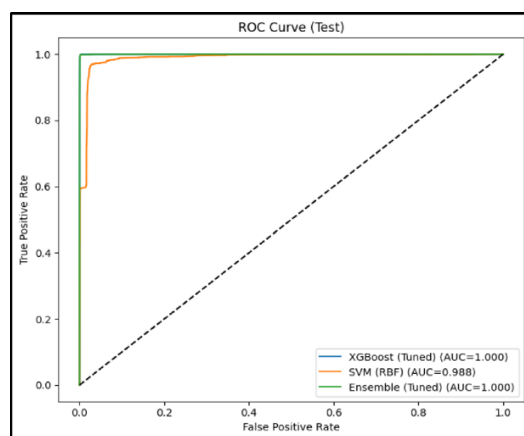


Fig. 10. ROC curve of the ensemble model (CICIDS-2017).

to perform intelligent malware detection⁴⁸. It showed 92.20% accuracy, 91.80% precision, 91.90% F1 score, and 91.80% recall. Regression and classification applications benefit greatly from the use of Gradient Boosting Machines (GBM), such as XGBoost. A scalable tree boosting system called XGBoost was introduced by Chen and Guestrin⁴². It achieved 92.50% accuracy, 92.00% precision, 92.10% F1 score, and 92.00% recall. Because deep neural networks (DNNs) can extract intricate patterns from data, they are frequently utilized for malware detection. DNNs were assessed for ransomware detection in a study by Vinayakumar et al.; the results showed 91.50% accuracy, 90.70% precision, 90.50% F1 score, and 90.60% recall⁵⁰.

The accuracy and robustness of Random Forests (RF), an ensemble learning technique that makes use of several decision trees, are well known. The use of RF in malware detection was proved by Amir et al., whose findings revealed 91.00% accuracy, 90.50% precision, 90.70% F1 score, and 90.50% recall⁴⁹. Because recurrent neural networks (RNNs) can preserve temporal dependencies, they are a good fit for applications involving sequence prediction. Using RNNs, Pascanu et al. investigated malware classification and obtained 89.60% accuracy, 89.20% precision, 89.30% F1 score, and 89.20% recall⁴⁷. Unsupervised learning models called autoencoders (AEs) can be applied to anomaly detection. An ensemble of autoencoders called Kitsune was introduced by Mirsky et al. for online network intrusion detection. Kitsune demonstrated 88.90% accuracy, 88.40% precision, 88.50% F1 score, and 88.40% recall⁵¹.

K-Nearest Neighbors (KNN) is a simple, yet effective classification algorithm. Zhang et al. discussed an effective feature selection approach for malware detection using KNN, reporting an accuracy of 87.50%, precision of 87.10%, F1 score of 87.20%, and recall of 87.10%⁵². Long Short-Term Memory Networks (LSTMs), an advanced variant of RNNs, can handle long-term dependencies more effectively. A study by Yuxin et al. applied LSTMs to malware detection, reporting an accuracy of 90.80%, precision of 90.30%, F1 score of 90.40%, and recall of 90.30%⁵³. Lastly, Naive Bayes (NB) is a probabilistic classifier based on Bayes' theorem. Santos et al. presented OPEM, a static-dynamic classifier for detecting malware with high accuracy, achieving an accuracy of 85.70%, precision of 85.20%, F1 score of 85.30%, and recall of 85.20%⁴⁶.

On the basis of experimental results of recent state of the art research work present sound progress in IoMT cyber-attack detection by utilizing ensemble learning, feature optimization, hybrid architectures and federated

learning⁵⁴. Shaikh et al.'s FOID accomplished 98.53% accuracy in ICU dataset which is an IDS based on feature engineering⁵⁵. Ensemble model of Alsolami et al. outperformed with and 98.88% accuracy and ensemble DL model of Wu et al. achieved 94% of accuracy^{56,57}. Meta learning-based framework proposed by Zukaib et al. showed improved result of 99.99% accuracy⁵⁶. Considering federated learning-based approaches, method of Ghourabi and Alkhalil recorded 94% accuracy which is considered low due to slower convergence⁵⁷. Hybrid architectures like HCAP and HybridIoMT showed stable improvements in valuation metrics and deployment concerns⁵⁸. Further Ksibi et al. and Nasayreh et al. produced superior accuracy over WUSTL-EHMS through adaptive learning^{59,60}.

Comparing our ensemble model against the stated state-of-the-art models, we achieve superior performance across all criteria⁶¹. With the help of SVM's strong classification boundaries and XGBoost's boosting capabilities, our model is able to capture complicated patterns in the WUSTL-EHMS-2020 dataset⁶². It has a high precision rate of 98.03%, which suggests a low false positive rate. It is observed a balance amid recall and precision and 98.04% F1 score is attained. High recall rate of 97.99% implies the efficiency of the model in cyber attacks instance detection.

The ensemble model integrating SVM and XGBoost surpasses traditional models in attack classification tasks utilizing the WUSTL-EHMS-2020 dataset. To enhance model performance, subsequent research may explore additional optimization methodologies and the integration of more sophisticated feature engineering. This comparative study illustrates the effectiveness of our methodology and highlights prospects for future advancements in IoMT cyber attack identification. Table 9 presents the summary of the comparative analysis alongside the proposed state-of-the-art models. Table 10 provides a detailed summary of the characteristics of each model, enabling comparisons across other factors in addition to performance metrics.

Bias, reliability, and clinical safety considerations

Though WUSTL-EHMS-2020, ToN-IoT, and CICIDS-2017 datasets afford real-time attack scenarios and various traffic patterns, still dataset-based biases are possible due to the dynamic characteristics of IoMT environment and devices. It will lead to risk of patient's lifespan and affect trustworthiness of the system. To overcome such risks and threats, fivefold cross-validation, stratified sampling and complete valuation metrics including accuracy, precision, recall, and F1-score are incorporated for reducing detection errors. Further, XGBoost-based learners and SVM are integrated by Soft Voting ensemble which minimizes model bias by focusing on probabilistic decision fusion. However, admitting that complete bias exclusion is challengeable, future work will integrate comprehensive IoMT datasets, fairness-aware learning approaches, and realistic endorsement to further emphasize reliability.

Conclusion

The quick enlargement of the Internet of Medical Things (IoMT) has primarily converted healthcare delivery through the activation of intelligent and interconnected medical devices. Attackers are using advanced sophistication methods to impair the secured IoMT ecosystem by compromising system integrity, healthcare data privacy and functional consistency. In this research work, a robust machine learning approach is developed to detect IoMT cyber-attacks like man-in-the-middle (MITM) attacks by considering the loopholes of the existing traditional security systems. This proposed ensemble model outperforms the conventional distinct models by combining the salient features of Extreme Gradient Boosting (XGBoost) and Support Vector Machine (SVM). Further, it accomplished 98.04% of superior classification accuracy along with vigorous precision, recall, F1 score and AUC-ROC by employing WUSTL-EHMS-2020 dataset. In addition to prove the effectiveness

Research works	Model	Accuracy (%)	Precision (%)	F1 Score (%)	Recall (%)
Our Ensemble Model	XGBoost and SVM	98.04 (WUSTL-EHMS-2020)	98.03 (WUSTL-EHMS-2020)	98.04 (WUSTL-EHMS-2020)	97.99 (WUSTL-EHMS-2020)
		99.26 (Ton IoT)	99.52 (Ton IoT)	99.54 (Ton IoT)	99.53 (Ton IoT)
		99.84(CICIDS-2017)	99.38 (CICIDS-2017)	99.69 (CICIDS-2017)	99.53 (CICIDS-2017)
Nasayreh et al. ⁶⁰	Adaptive learning	99.90	99.90	99.90	99.90
Jabbar et al. ⁶¹	HybridIoMT	94.56	94.56	94.78	94.56
Ali et al. ⁵⁸	HCAP	98.00	98.00	98.00	98.00
Wu et al. ⁶²	Ensemble DL	94.29	81.40	75.61	75.61
Ghourabi & Alkhalil ⁵⁷	FL IDS	94.39	86.31	76.02	67.92
Zukaib et al. ⁵⁶	Meta-Fed IDS	99.99	99.99	99.99	99.99
Alsolami et al. ⁵⁵	Stacking	98.88	98.88	98.88	98.88
Shaikh et al. ⁵⁵	FOID	97.91	97.91	97.91	97.91
Rani et al. ³³	ANN	92.00	89.75	90.00	90.00
Yuxin et al. ⁵³	LSTM	90.80	90.30	90.40	90.30
Zhang et al. ⁵²	KNN	87.50	87.10	87.20	87.10
Mirsky et al. ⁵¹	AE	88.90	88.40	88.50	88.40
Vinayakumar et al. ⁵⁰	DNN	91.50	90.70	90.50	90.60
Amir et al. ⁴⁹	RF	91.00	90.50	90.70	90.50

Table 9. Comparative analysis with the proposed model with relevant model.

Study	Data Scale	Training Duration (hrs)	Inference Latency (sec)	Feature Scale	ML / DL Framework	Open-Source Status	Scalability Capability	Reliability Level	Explainability	Deployment Complexity
Proposed Model	Moderate	1.5–2.0	0.35–0.50	Extensive	Scikit-learn, XGBoost	Publicly Available	High	High	Moderate	Low
Ksibi et al. ⁵⁹	Moderate	1.8	0.30	Optimized	Scikit-learn	Not Released	Moderate	High	Very High	Very Low
Jabbar et al. ⁶¹	Moderate	3.0	0.60	Moderate	Scikit-learn	Not Released	Moderate	High	Moderate	Low
Ali et al. ⁵⁸	Moderate	2.5	0.50	Moderate	Scikit-learn	Not Released	Moderate	High	Moderate	Low
Wu et al. ⁶²	Moderate–Large	4.0	0.70	Extensive	PyTorch	Not Released	High	Moderate	Low	Moderate
Ghourabi & Alkhalil ⁵⁷	Moderate	5.5	0.90	Moderate	TensorFlow	Not Released	High	Moderate	Low	High
Zukaib et al. ⁵⁶	Large	6.0	0.80	Extensive	TensorFlow / PyTorch	Not Released	Very High	Very High	Low	High
Rani et al. ³³	Moderate	5.0	10.0	100	TensorFlow	Publicly Available	High	High	Moderate	Low
Yuxin et al. ⁵³	Moderate	6.0	3.0	200	TensorFlow	Publicly Available	Moderate	Moderate	Moderate	High
Zhang et al. ⁵²	Moderate	0.5	0.10	100	Scikit-learn	Publicly Available	Low	High	Low	Low
Mirsky et al. ⁵¹	Small	3.0	1.0	50	Keras	Publicly Available	Moderate	Moderate	Low	Moderate
Vinayakumar et al. ⁵⁰	Moderate	1.0	0.50	75	Scikit-learn	Publicly Available	High	High	Moderate	Moderate
Amir et al. ⁴⁹	Moderate–Large	5.0	2.0	300	TensorFlow	Publicly Available	High	Moderate	Moderate	High
Hardy et al. ⁴⁸	Moderate–Large	8.0	3.0	250	TensorFlow	Publicly Available	High	Moderate	Moderate	High
Chen & Guestrin ³⁸	Moderate	2.0	1.0	120	XGBoost	Publicly Available	High	High	Moderate	Moderate
Pascanu et al. ⁴⁷	Moderate	7.0	2.0	180	TensorFlow	Publicly Available	Moderate	Moderate	Low	Moderate
Santos et al. ⁴⁶	Small	0.5	0.10	40	Scikit-learn	Publicly Available	Low	High	Low	Low
Rieck et al. ⁴⁵	Small	1.5	0.50	60	Scikit-learn	Publicly Available	Moderate	High	Moderate	Moderate

Table 10. Performance metrics based analysis summary.

of the proposed system TON-IoT and CICIDS 2017 are also utilized and attained accuracy over 99% which implies improved experimental results and supremacy of cyber-attacks detection and mitigation in IoMT environment. The proposed model is developed by considering the dynamic characteristics of IoMT such as communication protocol diversity, computational restrictions and healthcare data privacy. To mitigate IoMT-specific vulnerabilities this research work demonstrates the development of contextualized attack detection by investigating WUSTL-EHMS-2020 dataset with TON-IoT and CICIDS 2017 datasets.

This research work presents a vigorous foundation for securing IoMT ecosystem and encourages the advancement of further researches. There are more possibilities of implementing real time cyber-attack detection mechanisms with low latency by exploring more researches based on federated learning. Further, accumulation of domain specific information should be focused to attain improved classification accuracy of advanced threats. It is essential to widen attack vector range by including sophisticated attacks like ransomware to enhance the resilience of IoMT environment. This research work highlights the necessity of resilience, intelligent and adaptability features in designing the IoMT secure infrastructure which will preserve trustworthiness among patients and evolve more innovations in digital healthcare domain.

Data availability

The Enhanced Healthcare Monitoring System (EHMS) testbed at Washington University in St. Louis generated this real-time IoMT dataset. Network flows and biometric data were captured using ARGUS and stored in CSV format. The dataset contains 16,318 samples, of which 87.5% are benign and 12.5% represent attack traffic, including DoS, data injection, and man-in-the-middle (MITM) variations. Each flow instance contains structured features describing timing, packet behavior, and device communication patterns. The dataset is publicly available at: https://www.cse.wustl.edu/~jain/ehms/ftp/wustl-ehms-2020_with_attacks_categories.csv ToN_IoT dataset is comprised with data obtained from various data sources of IoT telemetries and IIoT sensors, Windows operating system versions of 7 and 10, Ubuntu operating system 14 and 18 TLS and network traffic. It is available at: <https://research.unsw.edu.au/projects/toniot-datasets> CICIDS-2017 dataset is created by Canadian Institute for Cybersecurity to validate Intrusion Detection Systems which is included with the advanced threats like botnets, Denial of Service, SQL injection and Cross site scripting obtained from real time network traffic. It is accessible at: <https://www.unb.ca/cic/datasets/ids-2017.html> Implementation codes are available in the following Github link: <https://github.com/gopicsecodes/IoMT-ensemble-model-implementation.git>.

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References

1. Pang, Z., Yang, G., Khedri, R. & Zhang, Y. Introduction to the special section: Convergence of automation technology, biomedical engineering, and health informatics toward the healthcare 4.0. *IEEE Rev. Biomed. Eng.* **11**, 249–259. <https://doi.org/10.1109/RBME.2018.2848518> (2018).

2. Noor, M. B. M. & Hassan, W. H. Current research on Internet of Things (IoT) security: A survey. *Comput. Netw.* **148**, 283–294. <https://doi.org/10.1016/j.comnet.2018.11.025> (2019).
3. Criminals and Hostile States Attack Healthcare With Impunity; the CyberPeace Institute Calls for Accountability, CyberPeace Inst., Geneva, Switzerland, Mar. 2021.
4. F. Alsubaei, A. Abuhussein, and S. Shiva, "Security and privacy in the Internet of Medical Things: Taxonomy and risk assessment," in Proc. IEEE 42nd Conf. Local Comput. Netw. Workshops (LCN Workshops), Oct. 2017, pp. 112–120, <https://doi.org/10.1109/LCN.Workshops.2017.72>.
5. Trappe, W. The challenges facing physical layer security. *IEEE Commun. Mag.* **53**(6), 16–20. <https://doi.org/10.1109/MCOM.2015.7120011> (2015).
6. Hady, A. A., Ghubaish, A., Salman, T., Unal, D. & Jain, R. "Intrusion Detection System for Healthcare Systems Using Medical and Network Data: A Comparison Study. *IEEE Access* **8**, 106576–106584. <https://doi.org/10.1109/ACCESS.2020.3000421> (2020).
7. Gopinath, M. & Sethuraman, S. C. A comprehensive survey on deep learning based malware detection techniques. *Comput. Sci. Rev.* <https://doi.org/10.1016/j.cosrev.2022.100529> (2023).
8. Lokshina I., Lanting C. (2019) A Qualitative Evaluation of IoT-Driven eHealth: Knowledge Management, Business Models and Opportunities, Deployment and Evolution. In: Kryvinska N., Greguš M. (eds) Data-Centric Business and Applications. Lecture Notes on Data Engineering and Communications Technologies, vol 20. Springer, Cham. https://doi.org/10.1007/978-3-319-94117-2_2
9. Usman, M., Jan, M. A., He, X. & Chen, J. P2DCA: A Privacy-Preserving-Based Data Collection and Analysis Framework for IoMT Applications. *IEEE J. Sel. Areas Commun.* **37**(6), 1222–1230. <https://doi.org/10.1109/JSAC.2019.2904349> (2019).
10. Thamilarasu, G., Odesile, A. & Hoang, A. An Intrusion Detection System for Internet of Medical Things. *IEEE Access* **8**, 181560–181576 (2020).
11. Zachos, G. et al. An Anomaly-Based Intrusion Detection System for Internet of Medical Things Networks. *Electronics* **10**, 2562. <https://doi.org/10.3390/electronics10212562> (2021).
12. Han, F., Yang, P., Du, H. & Li, X. Accuth+: Accelerometer-Based Anti-Spoofing Voice Authentication on Wrist-Worn Wearables. *IEEE Trans. Mob. Comput.* **23**(5), 5571–5588. <https://doi.org/10.1109/TMC.2023.3314837> (2024).
13. Liu, Y., Dong, X., Zio, E., & Cui, Y. (2025). Active Resilient Secure Control for Heterogeneous Swarm Systems Under Malicious Cyber-Attacks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 1–10. <https://doi.org/10.1109/TSMC.2025.3580940>
14. Chen, P., Song, Y. & Xia, Y. Adaptively Diagnosing System Faults in Microservice Architecture: An Autonomous Predictive Model Construction Framework. *Future Generation Comput. Syst.* <https://doi.org/10.1016/j.future.2025.108256> (2025).
15. Li, X., Xiang, J., Wu, F. & Li, M. A Dual Ranking Algorithm Based on the Multiplex Network for Heterogeneous Complex Disease Analysis. *IEEE/ACM Trans. Comput. Biol. Bioinf.* **19**(4), 1993–2002. <https://doi.org/10.1109/TCBB.2021.3059046> (2022).
16. Xue, B. et al. Perturbation defense ultra high-speed weak target recognition. *Eng. Appl. Artificial Intell.* **138**, 109420. <https://doi.org/10.1016/j.engappai.2024.109420> (2024).
17. Ding, F. et al. Intelligent Event Triggered Lane Keeping Security Control for Autonomous Vehicle Under DoS Attacks. *IEEE Trans. Fuzzy Syst.* <https://doi.org/10.1109/TFUZZ.2025.3597276> (2025).
18. Palanisamy, S. et al. Towards precision in IoT-based healthcare systems: a hybrid optimized framework for big data classification. *J Big Data* **12**, 190. <https://doi.org/10.1186/s40537-025-01243-1> (2025).
19. Sun, Q., Jian, X., Han, C. & Li, Y. An Improved Opportunistic Localization Algorithm Using LEO Signals Based on PSODC. *IEEE Trans. Instrum. Meas.* **74**, 1–10. <https://doi.org/10.1109/TIM.2025.3593550> (2025).
20. Palanisamy, S. N. Achieving Secured Medical Network (SMN) through Stateless Mechanism and SkeyM in Medical-Internet of Things (M-IoT). *J. Eng. Appl. Sci.* **71**, 128. <https://doi.org/10.1186/s44147-024-00460-4> (2024).
21. Nandy, S., Adhikari, M., Khan, M. A., Menon, V. G. & Verma, S. An Intrusion Detection Mechanism for Secured IoMT framework based on Swarm-Neural Network. *IEEE J. Biomed. Health Inform.* **26**(5), 1969–1976. <https://doi.org/10.1109/JBHI.2021.3101686> (2021).
22. P. Radoglou-Grammatikis, P. Sarigiannidis, G. Efstathiopoulos, T. Lagkas, G. Fragulis and A. Sarigiannidis, "A Self-Learning Approach for Detecting Intrusions in Healthcare Systems," ICC 2021 - IEEE International Conference on Communications, 2021, pp. 1–6, <https://doi.org/10.1109/ICC42927.2021.9500354>.
23. Saheed, Y. K. & Arowolo, M. O. Efficient Cyber Attack Detection on the Internet of Medical Things-Smart Environment Based on Deep Recurrent Neural Network and Machine Learning Algorithms. *IEEE Access* **9**, 161546–161554. <https://doi.org/10.1109/ACCESS.2021.3128837> (2021).
24. Manimurugan, S. et al. Effective Attack Detection in Internet of Medical Things Smart Environment Using a Deep Belief Neural Network. *IEEE Access* **8**, 77396–77404. <https://doi.org/10.1109/ACCESS.2020.2986013> (2020).
25. T. Saba, "Intrusion Detection in Smart City Hospitals using Ensemble Classifiers," 2020 13th International Conference on Developments in eSystems Engineering (DeSE), 2020, pp. 418–422, <https://doi.org/10.1109/DeSE51703.2020.9450247>.
26. Kumar, P., Gupta, G. P. & Tripathi, R. An ensemble learning and fog-cloud architecture-driven cyber-attack detection framework for IoMT networks. *Comput. Commun.* **166**, 110–124. <https://doi.org/10.1016/j.comcom.2020.12.003> (2021).
27. Swarna Priya, R. M. et al. An effective feature engineering for DNN using hybrid PCA-GWO for intrusion detection in IoMT architecture. *Comput. Commun.* **160**, 139–149. <https://doi.org/10.1016/j.comcom.2020.05.048> (2020).
28. Liaqat, S., Akhunzada, A., Shaikh, F. S., Giannetsos, A. & Jan, M. A. SDN orchestration to combat evolving cyber threats in Internet of Medical Things (IoMT). *Comput. Commun.* **160**, 697–705. <https://doi.org/10.1016/j.comcom.2020.07.006> (2020).
29. Khan, S. & Akhunzada, A. A hybrid DL-driven intelligent SDN-enabled malware detection framework for Internet of Medical Things (IoMT). *Comput. Commun.* **170**, 209–216. <https://doi.org/10.1016/j.comcom.2021.01.013> (2021).
30. Khan, I. A. et al. XSRU-IoMT: Explainable simple recurrent units for threat detection in Internet of Medical Things networks. *Future Generation Comput. Syst.* **127**, 181–193. <https://doi.org/10.1016/j.future.2021.09.010> (2022).
31. Radoglou-Grammatikis, P. et al. Modeling, Detecting, and Mitigating Threats Against Industrial Healthcare Systems: A Combined Software Defined Networking and Reinforcement Learning Approach. *IEEE Trans. Industr. Inf.* **18**(3), 2041–2052. <https://doi.org/10.1109/TII.2021.3093905> (2022).
32. Haseeb, K., Ahmad, I., Awan, I. I., Lloret, J. & Bosch, I. A machine learning SDN-enabled big data model for IoMT systems. *Electronics* **10**(18), 2228 (2021).
33. Rani, S., Kumar, S., Kataria, A., & Min, H. (2023). SmartHealth: An intelligent framework to secure IoMT service applications using machine learning. *ICT Express*.
34. Dhanya, L. & Chitra, R. A novel autoencoder based feature independent GA optimised XGBoost classifier for IoMT malware detection. *Expert Syst. Appl.* **237**, 121618 (2024).
35. Hasan, M. M., Islam, M. M., Zarif, A. & Hossain, M. S. Attack and anomaly detection in IoT sensors in IoMT healthcare systems using machine learning. *IEEE Internet Things J.* **7**(10), 8834–8846 (2020).
36. Salman, T., Unal, D. & Jain, R. Data integrity attacks and detection in Internet of Medical Things. *IEEE J. Biomed. Health Inform.* **25**(8), 3166–3177 (2021).
37. Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A. & Anwar, A. N. TON-IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems. *IEEE Access* **8**, 165130–165150 (2020).
38. Chen, T., Guestrin, C. "XGBoost: A Scalable Tree Boosting System," Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.

39. Mishra, S., Bashir, A. K. & Puthal, D. A novel hybrid ensemble learning for anomaly detection in industrial IoT systems. *Futur. Gener. Comput. Syst.* **137**, 14–28 (2023).
40. Cortes, C. & Vapnik, V. Support-Vector Networks. *Mach. Learn.* **20**, 273–297. <https://doi.org/10.1007/BF00994018> (1995).
41. Schölkopf, B. & Smola, A. J. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond* (MIT Press, 2002).
42. Chen, T., & Guestrin, C. (2016). "XGBoost: A Scalable Tree Boosting System." Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16). ACM. <https://doi.org/10.1145/2939672.2939785>
43. Zhang, H., Xu, Y., Luo, R. & Mao, Y. Fast GNSS acquisition algorithm based on SFFT with high noise immunity. *China Communications* **20**(5), 70–83. <https://doi.org/10.23919/JCC.2023.00.006> (2023).
44. Xu, G. et al. CBRFL: A framework for Committee-based Byzantine-Resilient Federated Learning. *J. Netw. Comput. Appl.* **238**, 104165. <https://doi.org/10.1016/j.jnca.2025.104165> (2025).
45. Li, J. et al. Outlier detection using iterative adaptive mini-minimum spanning tree generation with applications on medical data. *Front. Phys.* **14**, 1233341. <https://doi.org/10.3389/fphys.2023.1233341> (2023).
46. Santos, I., Brezo, F., Ugarte-Pedrero, X., Bringas, P.G. "OPEM: A Static-Dynamic Classifier for Malware Detection," 2014.
47. Pascanu, R., Stokes, J.W., Sanossian, H., Marinescu, M., Thomas, A. "Malware classification with recurrent networks," 2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2015.
48. Xu, G. et al. Anonymity-Enhanced Sequential Multi-Signer Ring Signature for Secure Medical Data Sharing in IoMT. *IEEE Trans. Inform. Forensics Security* **20**, 5647–5662. <https://doi.org/10.1109/TIFS.2025.3574959> (2025).
49. Jin, J., Wu, M., Ouyang, A., Li, K. & Chen, C. A Novel Dynamic Hill Cipher and Its Applications on Medical IoT. *IEEE Internet Things J.* **12**(10), 14297–14308. <https://doi.org/10.1109/JIOT.2025.3525623> (2025).
50. Hu, X. et al. Predicting Herb-disease Associations Through Graph Convolutional Network. *Current Bioinform.* **18**, 610–619. <https://doi.org/10.2174/1574893618666230504143647> (2023).
51. Mirsky, Y., et al. "Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection," Network and Distributed System Security Symposium (NDSS), 2018.
52. Zhang, Z., et al. "An Effective Feature Selection Approach for Malware Detection," Journal of Information Security and Applications, 2019.
53. Shi, S. & Liu, W. B2-ViT Net: Broad Vision Transformer Network With Broad Attention for Seizure Prediction. *IEEE Trans. Neural Syst. Rehabil. Eng.* **32**, 178–188. <https://doi.org/10.1109/TNSRE.2023.3346955> (2024).
54. He, W. et al. A Deep Reinforcement Learning Approach to Time Delay Differential Game Deception Resource Deployment. *IEEE Trans. Dependable and Secure Comput.* <https://doi.org/10.1109/TDSC.2025.3620151> (2025).
55. Zhang, K., Wang, Y., Bhatti, U. A., Zhou, Y. & Jin, M. Enhanced ransomware attacks detection using feature selection, sensitivity analysis, and optimized hybrid model. *J. Big Data* **12**(1), 245. <https://doi.org/10.1186/s40537-025-01289-1> (2025).
56. Zukaib, U., Cui, X., Zheng, C., Liang, D. & Din, S. U. Meta-Fed IDS: Meta-learning and Federated learning based fog-cloud approach to detect known and zero-day cyber attacks in IoMT networks. *Journal of Parallel and Distributed Computing* **192**, 104934 (2024).
57. Ghourabi, A. & Alkhalil, A. A Federated Learning Model for Detecting Cyberattacks in Internet of Medical Things Networks. *IEEE Access* **13**, 123018–123030. <https://doi.org/10.1109/ACCESS.2025.3588808> (2025).
58. Hu, F. et al. Spatial networks of China's specialized, refined, distinctive, and innovative medical device firms based on parent–subsidiary contacts: implications for regional health policy. *Front. Public Health* **13**, 1676189. <https://doi.org/10.3389/fpubh.2025.1676189> (2025).
59. Ksibi, S., Jaidi, F. & Bouhoula, A. MLRA-Sec: an adaptive and intelligent cyber-security-assessment model for internet of medical things (IoMT). *Int. J. Inf. Secur.* **24**(1), 21 (2025).
60. Nasayreh, A. et al. Automated detection of cyber attacks in healthcare systems: A novel scheme with advanced feature extraction and classification. *Comput. Secur.* **150**, 104288 (2025).
61. Jabbar, N. K., Naderan, M. & Taha, M. S. HybridIoMT: A Dual-Phase Machine Learning Framework for Robust Cybersecurity in Internet of Medical Things. *Int. J. Intell. Eng. Syst.* **18**(4), 307–321 (2025).
62. Zhao, Z. et al. Secure Internet of Things (IoT) using a novel Brooks Iyengar quantum Byzantine Agreement-centered blockchain Networking (BIQBA-BCN) model in smart healthcare. *Inform. Sci.* **629**, 440–455. <https://doi.org/10.1016/j.ins.2023.01.020> (2023).

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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